

PROGETTO FINALE_W24D4_LAB DI SICUREZZA CON AD E SIEM

Obiettivo

Realizzare un ambiente virtualizzato con un dominio Active Directory monitorato da un SIEM, simulando almeno un evento di sicurezza rilevabile.

Infrastruttura richiesta:

•Macchina 1 — Domain Controller

Sistema operativo: Windows Server (2019 o 2022)

Ruolo: Active Directory Domain Services (AD DS) + DNS

Deve essere promossa a Domain Controller di un dominio a scelta (es. lab.local)

Almeno un utente di dominio creato (oltre all'Administrator)

•Macchina 2 — Client

Sistema operativo: Windows 10 o 11

Deve essere aggiunta al dominio

Accesso con l'utente di dominio creato sul DC

•Macchina 3 — SIEM

Sistema operativo: Linux (consigliato) o Windows

SIEM a scelta (es. Splunk Free, Wazuh, Elastic/ELK, Graylog...)

Deve ricevere i log dal Domain Controller e dal client

-Configurazione del forwarding

I log devono essere inviati al SIEM

-Simulazione di un evento di sicurezza

ESECUZIONE:

Infrastruttura del laboratorio virtuale:

Windows Server 2022 > AD DS + DNS abilitato

Windows 10 pro > Client da aggiungere al dominio lab.local

Kali linux > “attaccante”

SIEM Splunk > vm Ubuntu

WINDOWS SERVER 2022

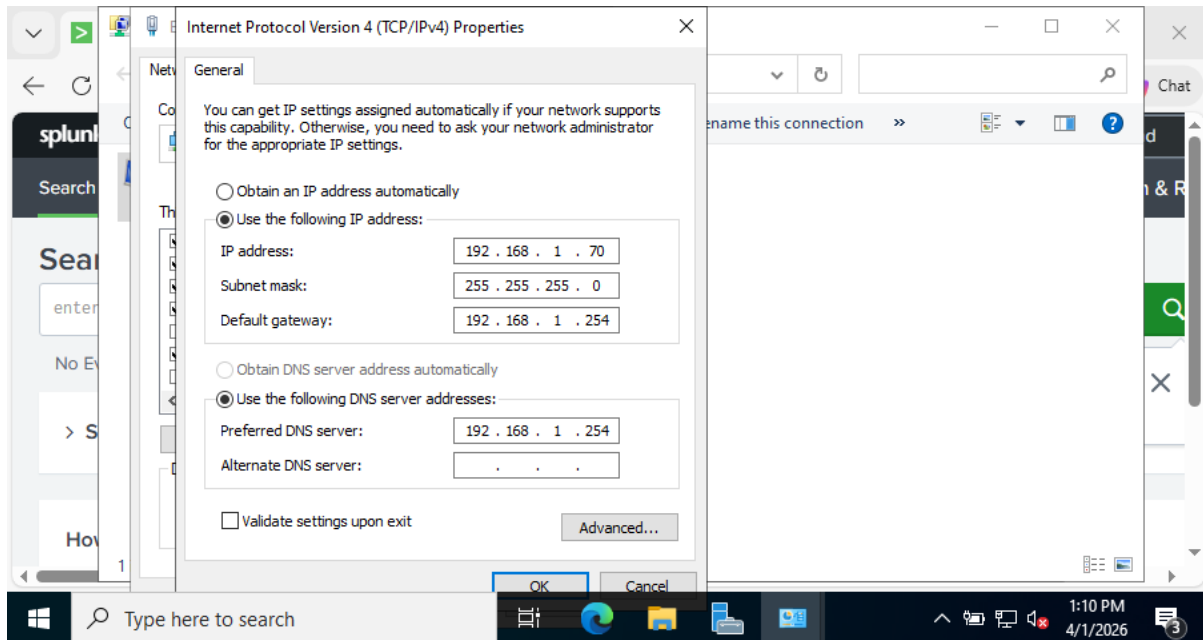
Nella sezione server manager si va ad effettuare l'installazione di una AD che gestisce tutta l'infrastruttura IT

Per accedere a windows server si utilizza il client windows 10 con un dominio standard.

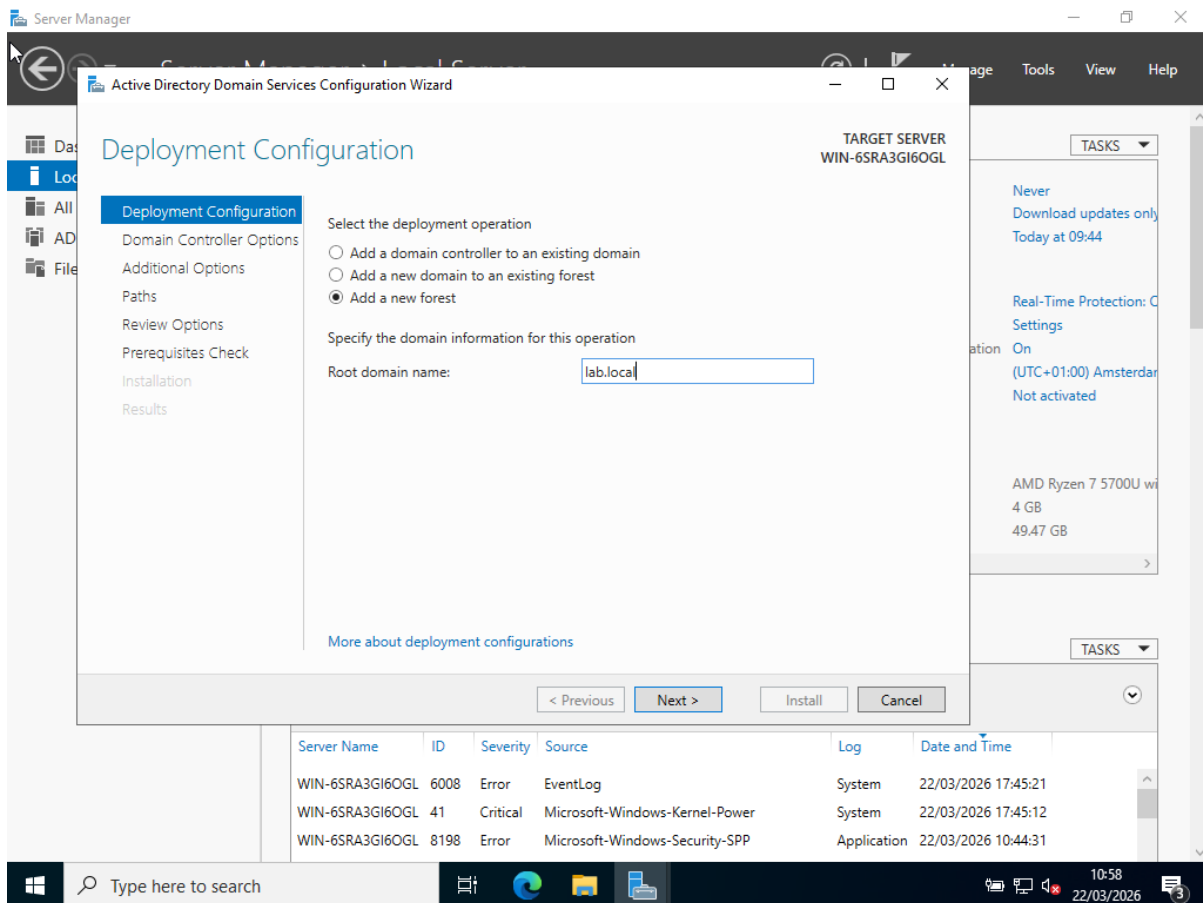
CONFIGURAZIONE IP STATICO E CONFIGURAZIONE SERVIZIO DOMAIN CONTROLLER

ip statico attribuito alla macchina 1 Windows Server per AD : 192.168.1.70

Gateway:192.168.1.254; DNS:192.168.1.254



CONFIGURAZIONE DELLA FORESTA

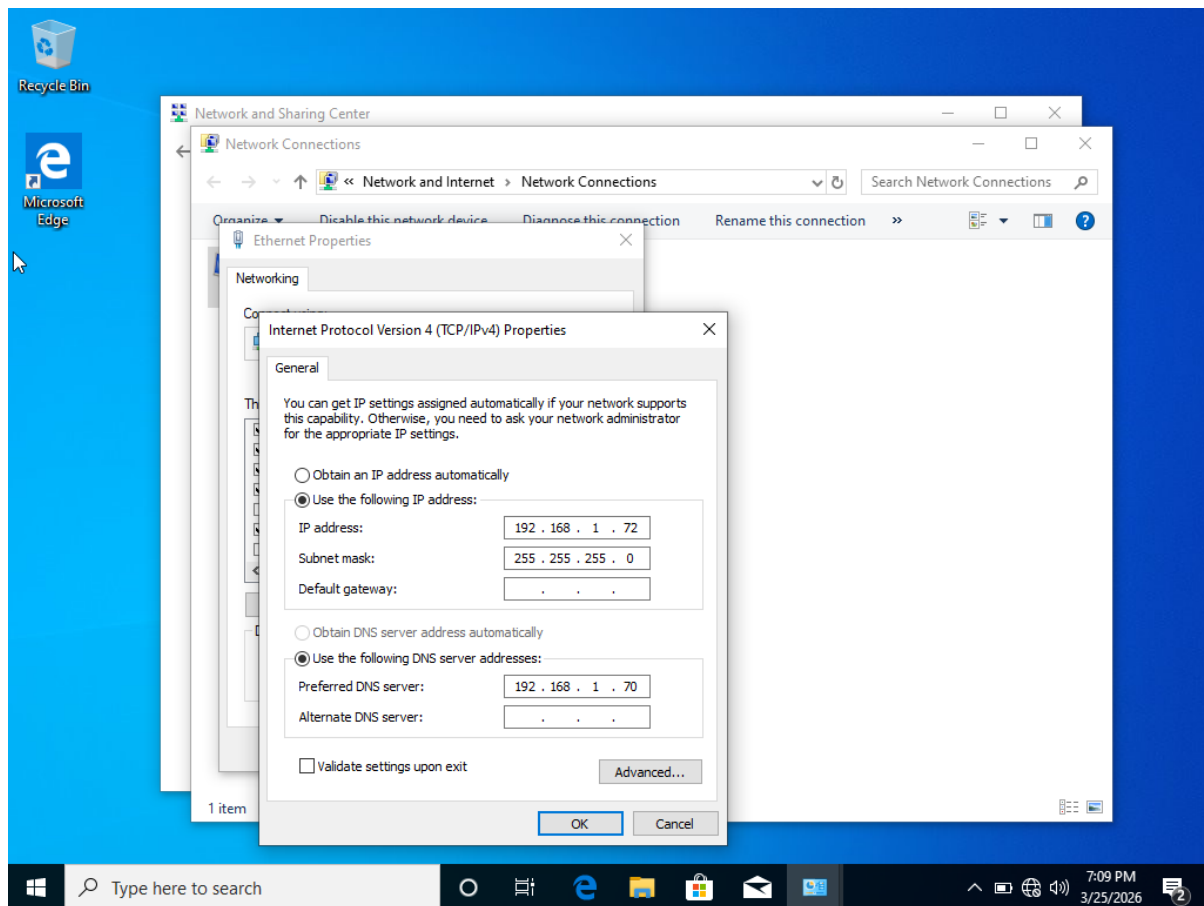


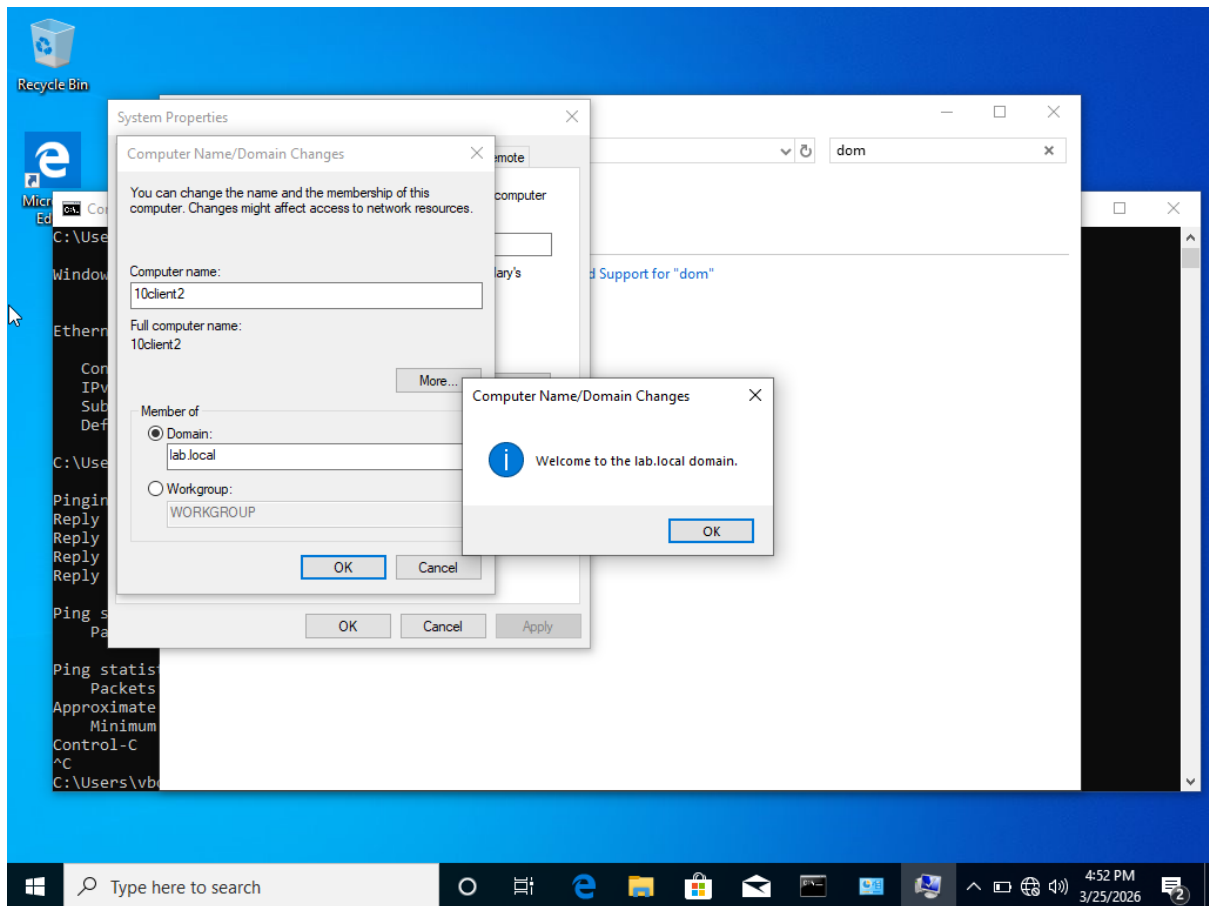
Il dominio LAB.LOCAL è l' administrator.

CONFIGURAZIONE CLIENT WINDOWS 10

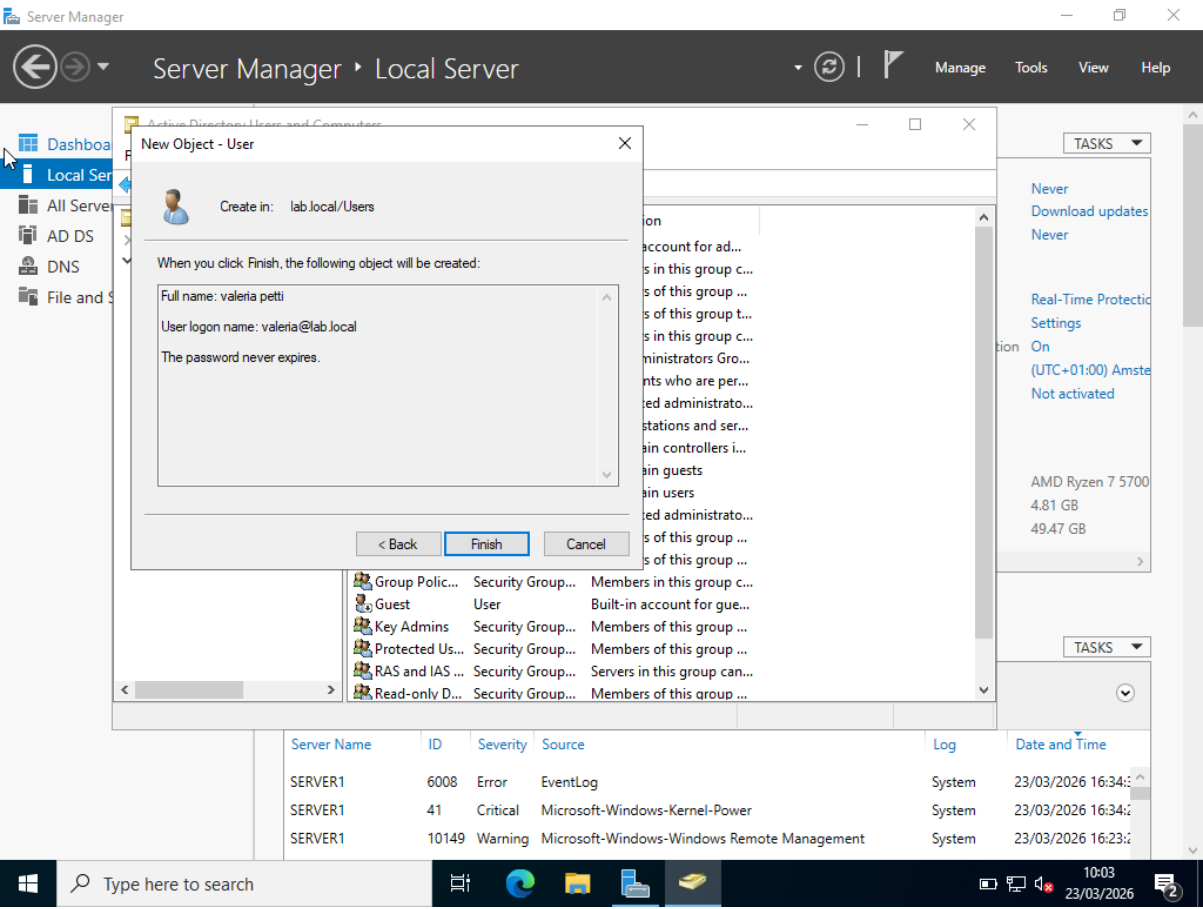
Il client per comunicare con il server utilizzerà il dns quindi l'ip del domain controller in quanto deve essere in grado di controllare i vari client appartenenti all'infrastruttura. Con una prova per verificare la comunicazione tra client e server ci si accerta che le macchine comunichino tra loro.

-in fase di configurazione è stato individuato un errore in quanto automaticamente la macchina client lasciava il flag anche sull'ipv6 -





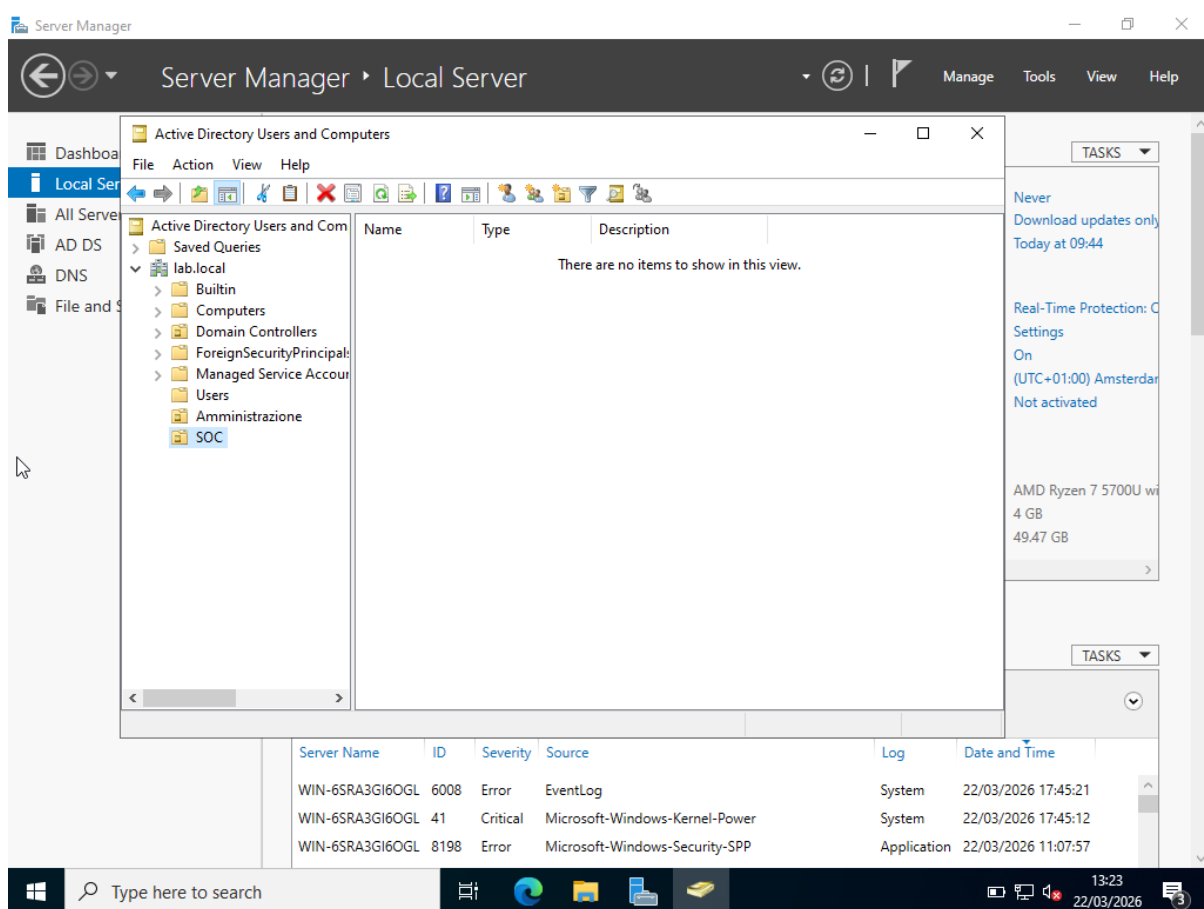
CREAZIONE UTENTE SU WINDOWS SERVER 2022 CHE FA PARTE DELLA FORESTA



CREAZIONE DI UNA UNITÀ ORGANIZZATIVA PER LA SIMULAZIONE DI UNA WORKSTATION

Come primo step bisogna crea una Organizational Unit denominata SOC con all'interno il gruppo SOC ADMINISTRATION che comprende due utenti i quali hanno dei permessi attribuiti dalle GPO.

Relativamente alla password è sempre consigliabile in un contesto aziendale scegliere un rinnovo al primo accesso.



Server Manager

Server Manager ▶ Local Server

ManageToolsViewHelp

Dashboard

Local Server

All Servers

AD DS

DNS

File and Storage Services

Active Directory Users and Computers

FileActionViewHelp

Active Directory Users and Computers

Saved Queries

lab.local

Builtin

Computers

Domain Controllers

ForeignSecurityPrincipals

Managed Service Accounts

Users

Amministrazione

SOC

Name	Type	Description
SOC ADMIN...	Security Group...	
soc admins	User	

TASKS

Never

Download updates only

Today at 09:44

Real-Time Protection: C

Settings

On

(UTC+01:00) Amsterdam

Not activated

AMD Ryzen 7 5700U wi

4 GB

49.47 GB

TASKS

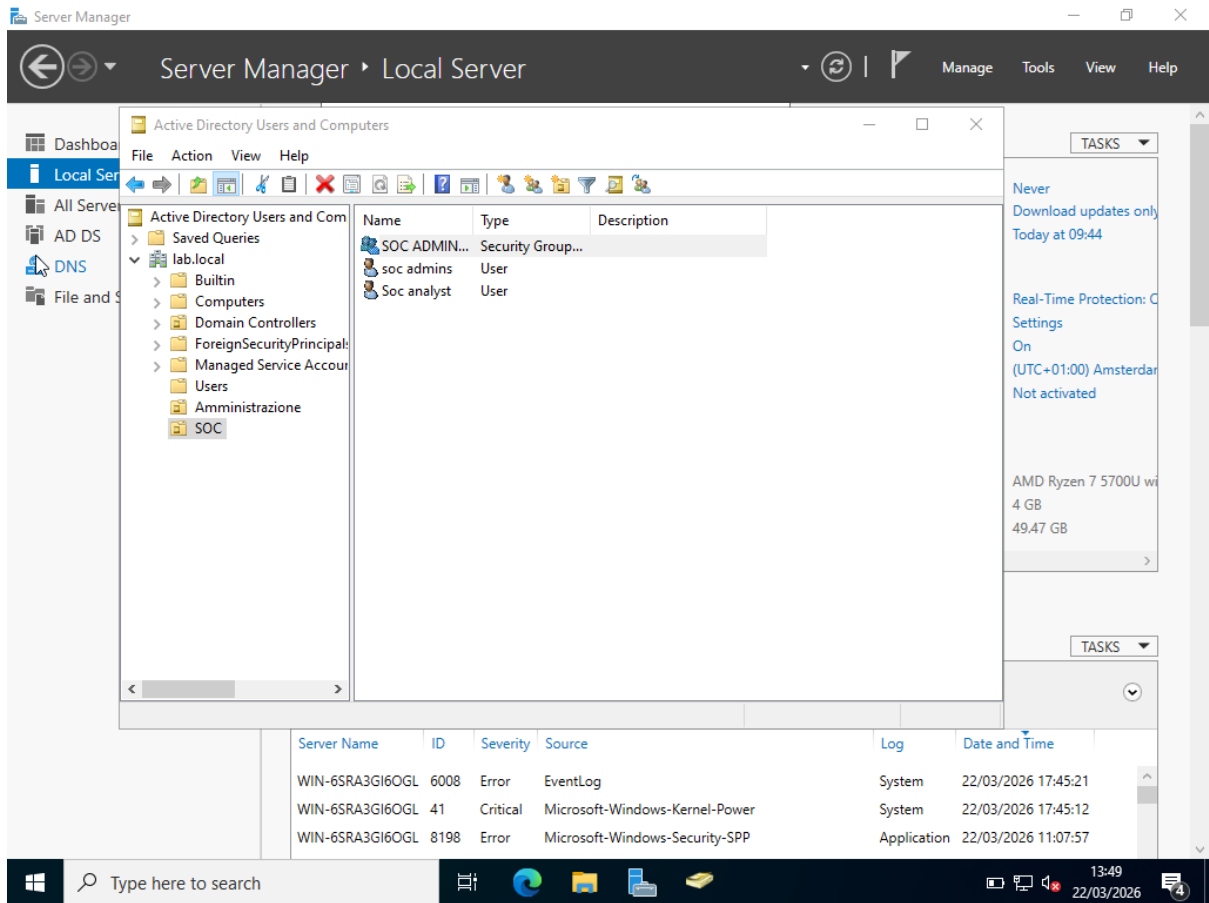
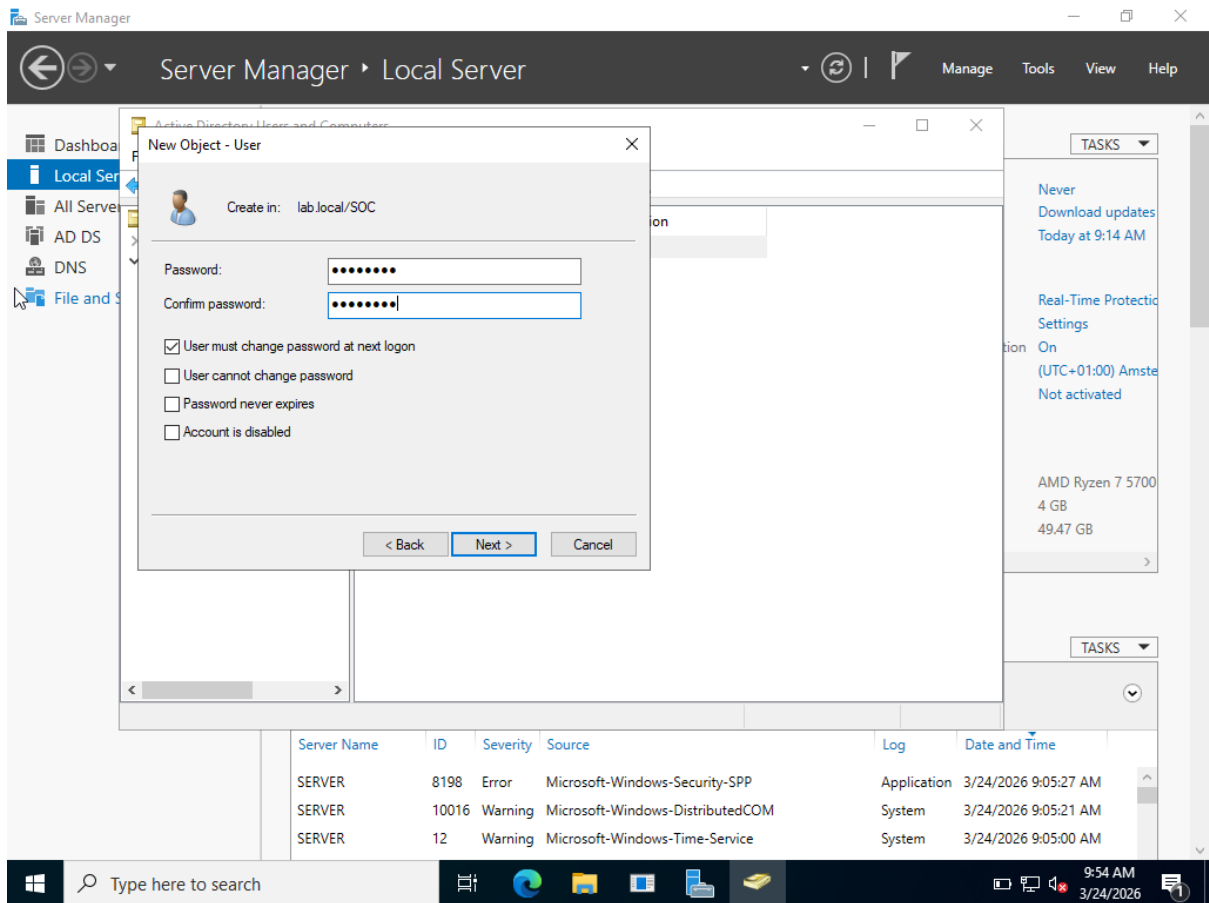
Server Name	ID	Severity	Source	Log	Date and Time
WIN-6SRA3GI6OGL	6008	Error	EventLog	System	22/03/2026 17:45:21
WIN-6SRA3GI6OGL	41	Critical	Microsoft-Windows-Kernel-Power	System	22/03/2026 17:45:12
WIN-6SRA3GI6OGL	8198	Error	Microsoft-Windows-Security-SPP	Application	22/03/2026 11:07:57

Type here to search

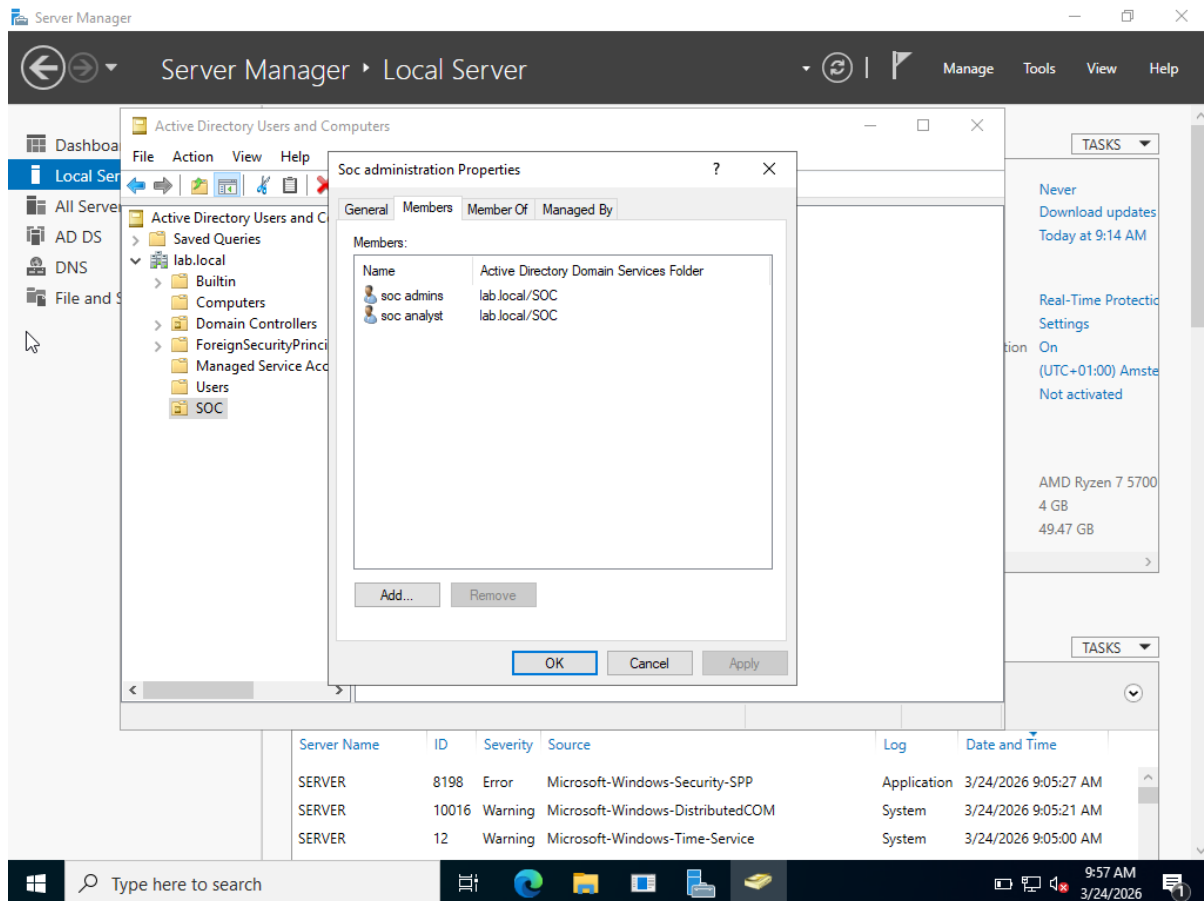
13:35

22/03/2026

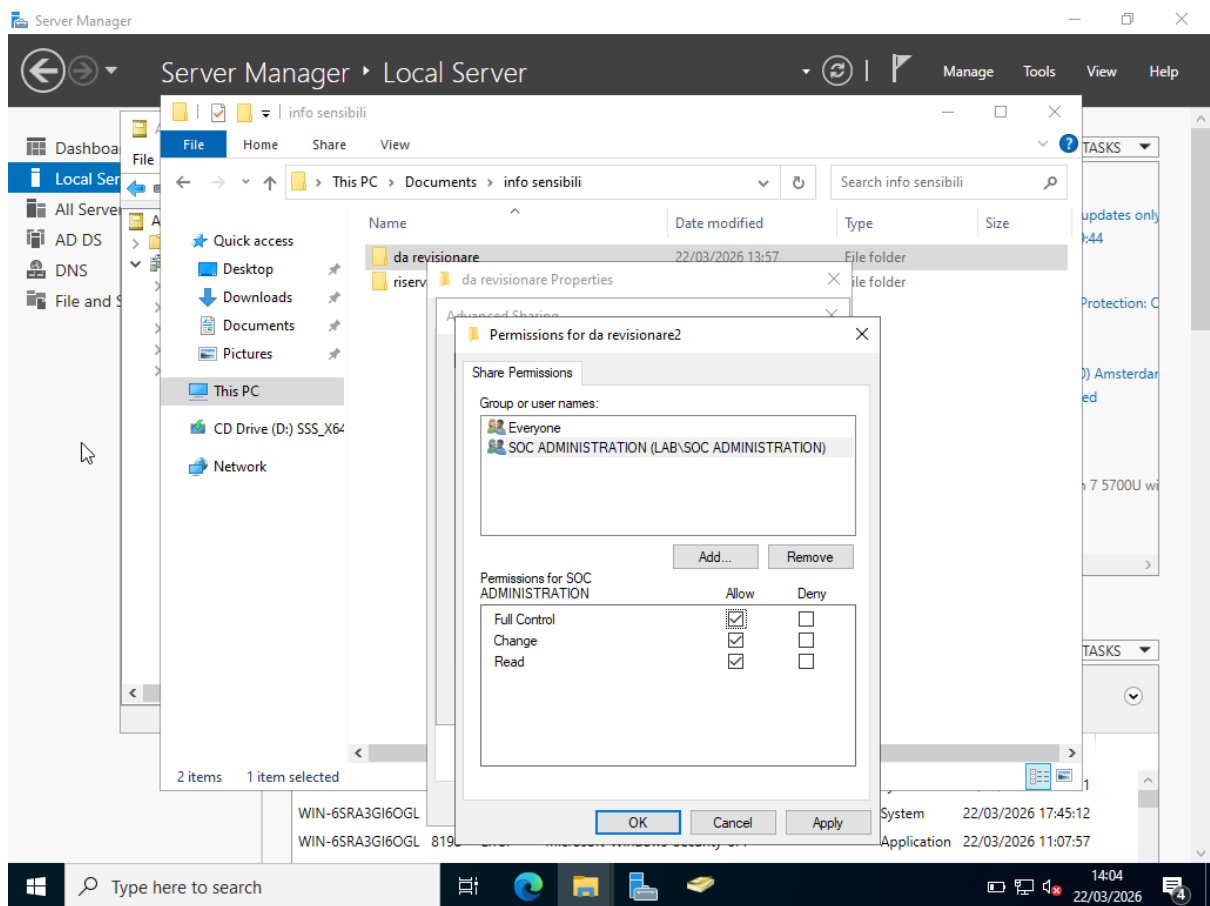
4



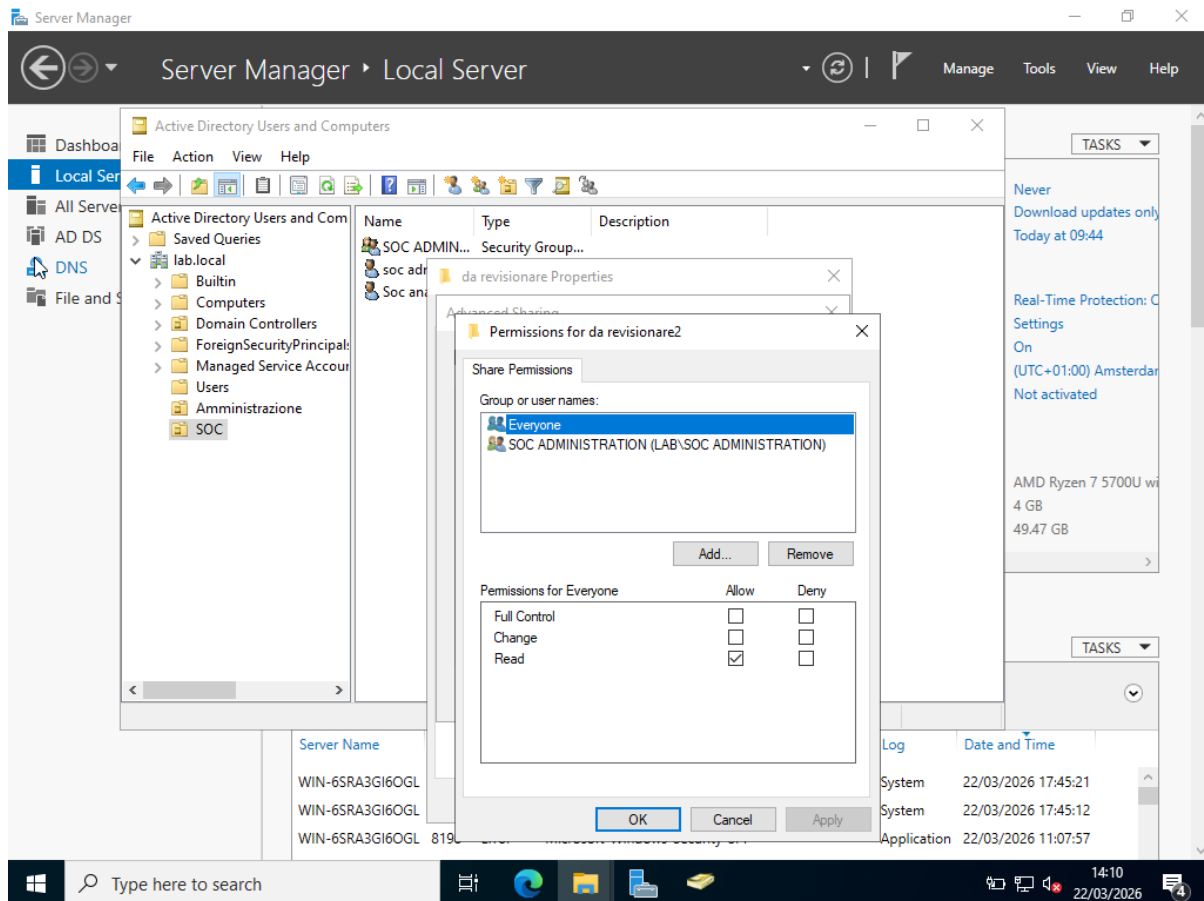
I due utenti all'interno del gruppo soc administration sono il soc admins e il soc analyst.



I permessi attribuiti al gruppo sono quelli di lettura, scrittura ed esecuzione

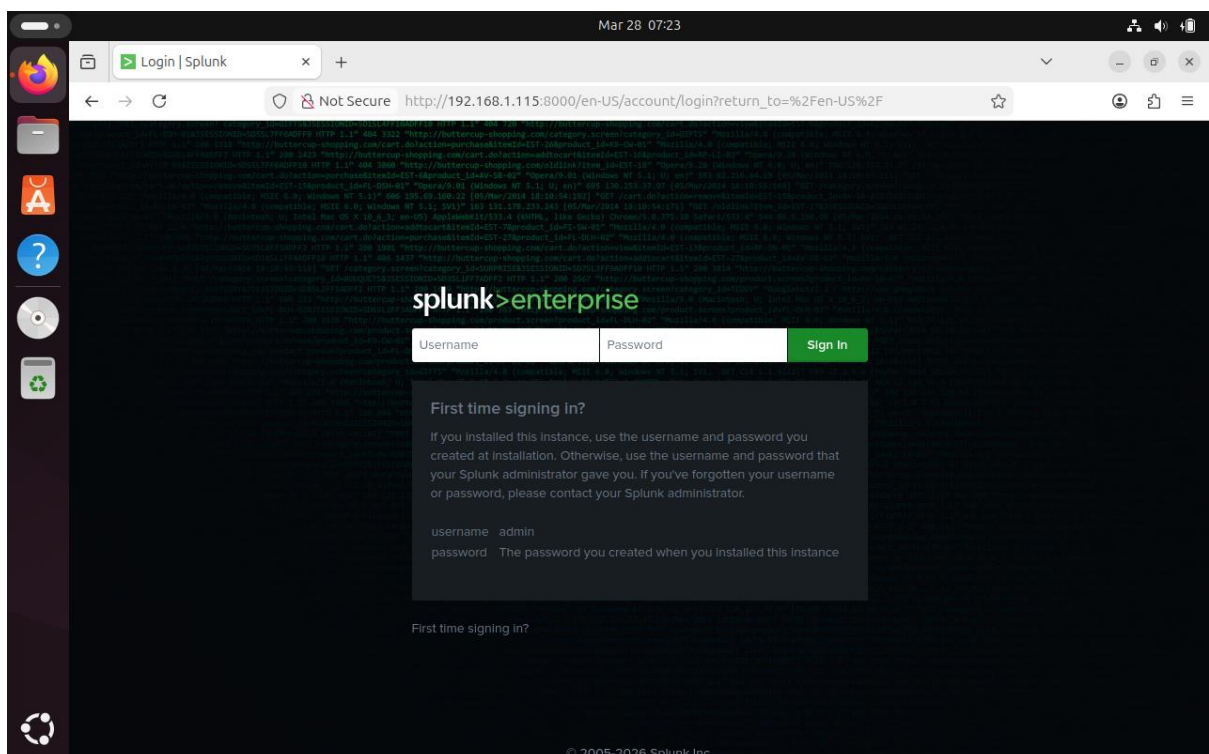
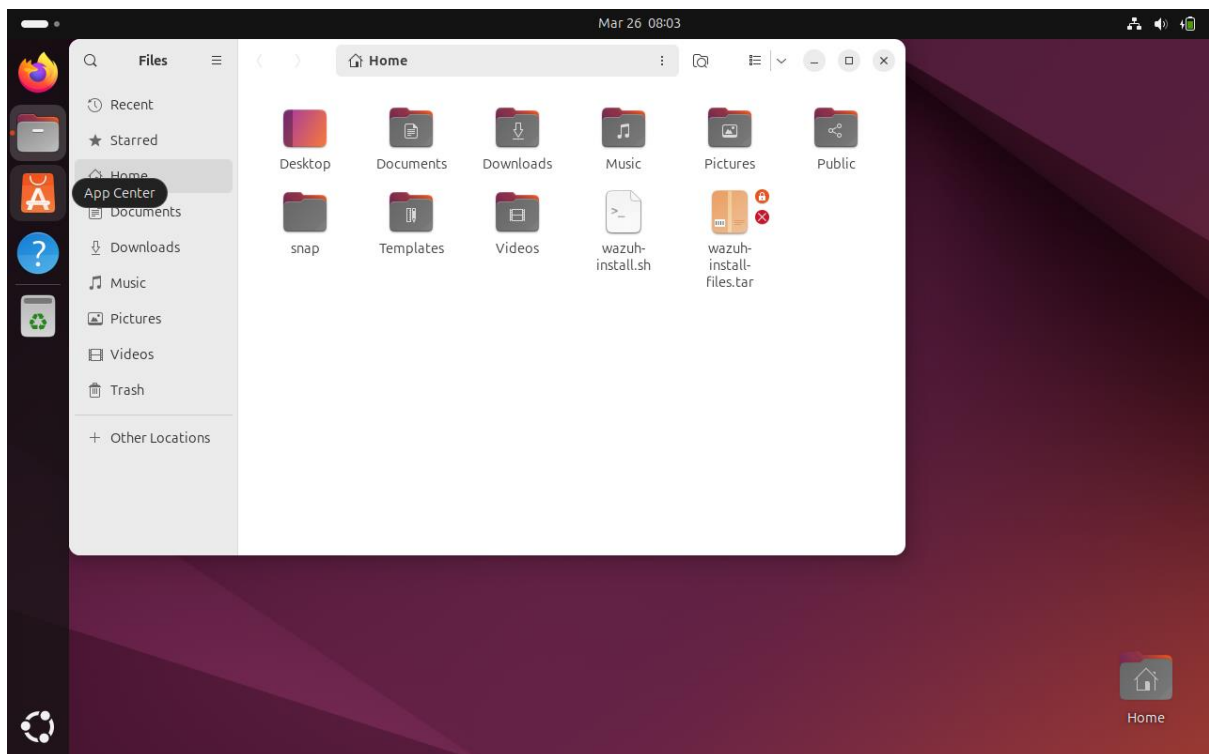


Il resto degli utenti ha solo permessi di lettura

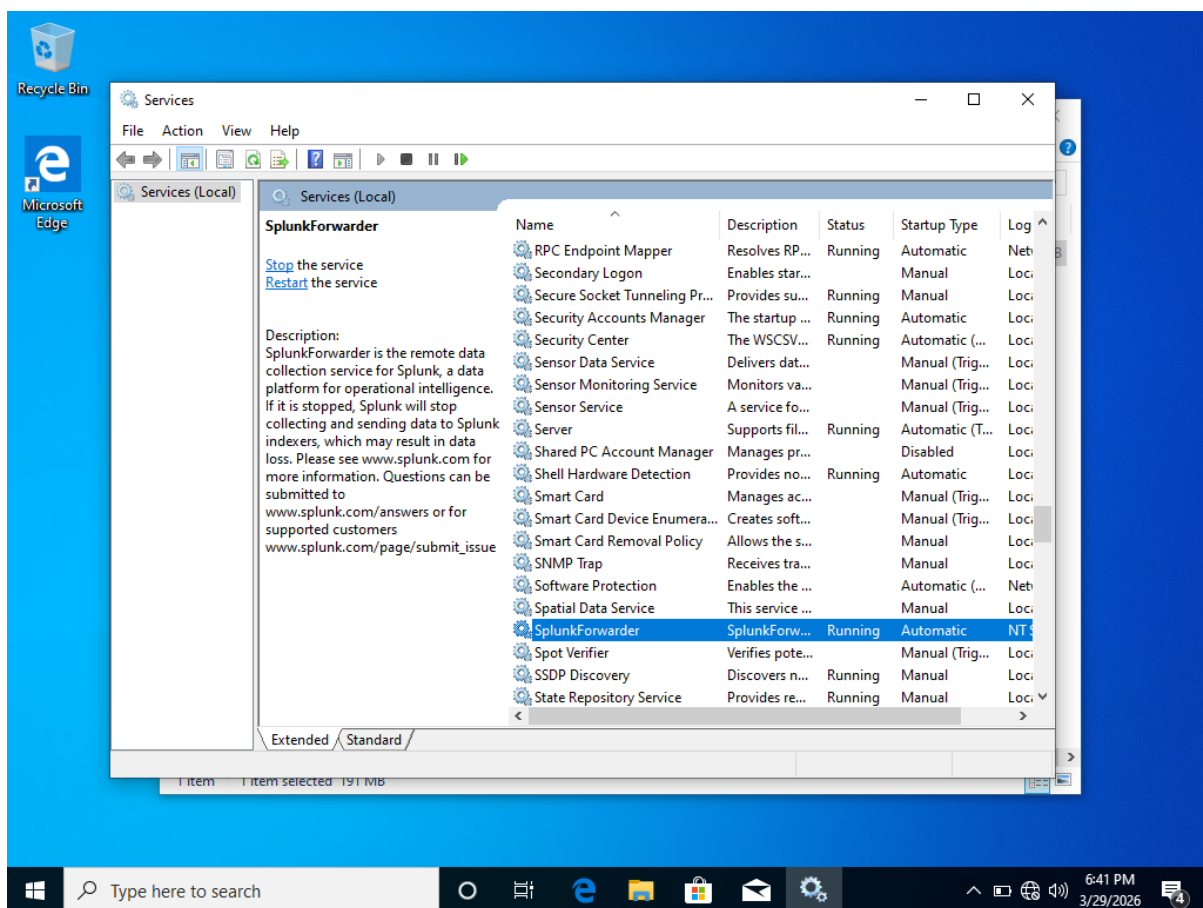
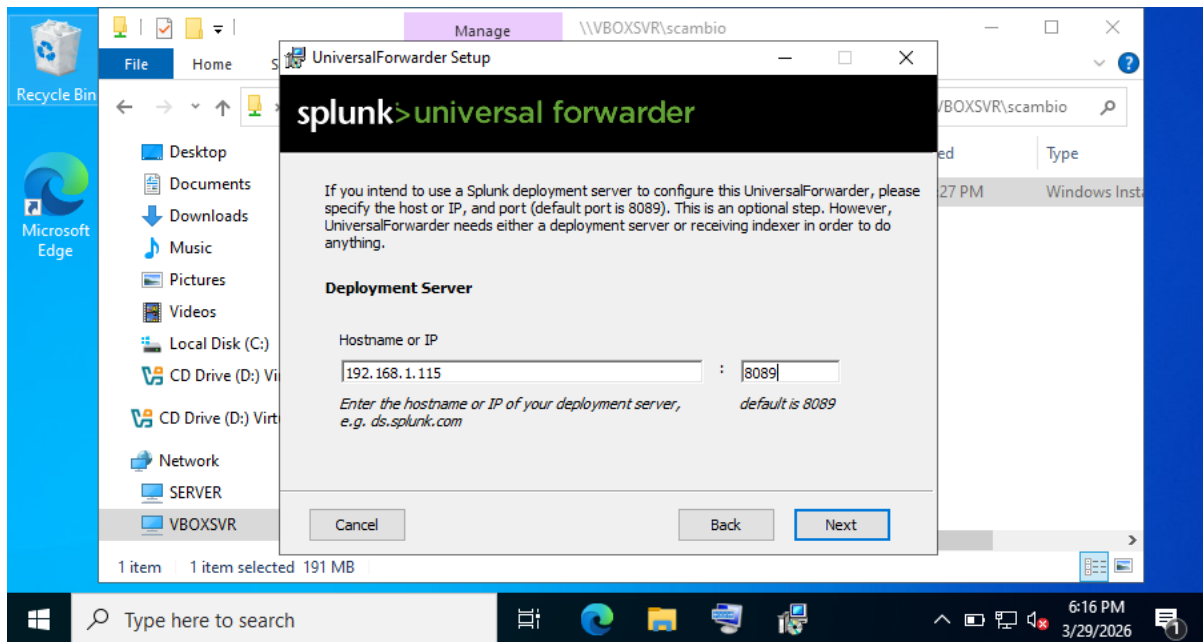


SU UNA VM UBUNTU SI VA AD INSTALLARE IL SIEM SPLUNK CHE TIENE TRACCA DEI LOG

-è stato utilizzato il servizio ssh per l'installazione di splunk sulla vm ubuntu . Wazuh ha invece richiesto diverse ore per terminare ed iniziare il funzionamento -

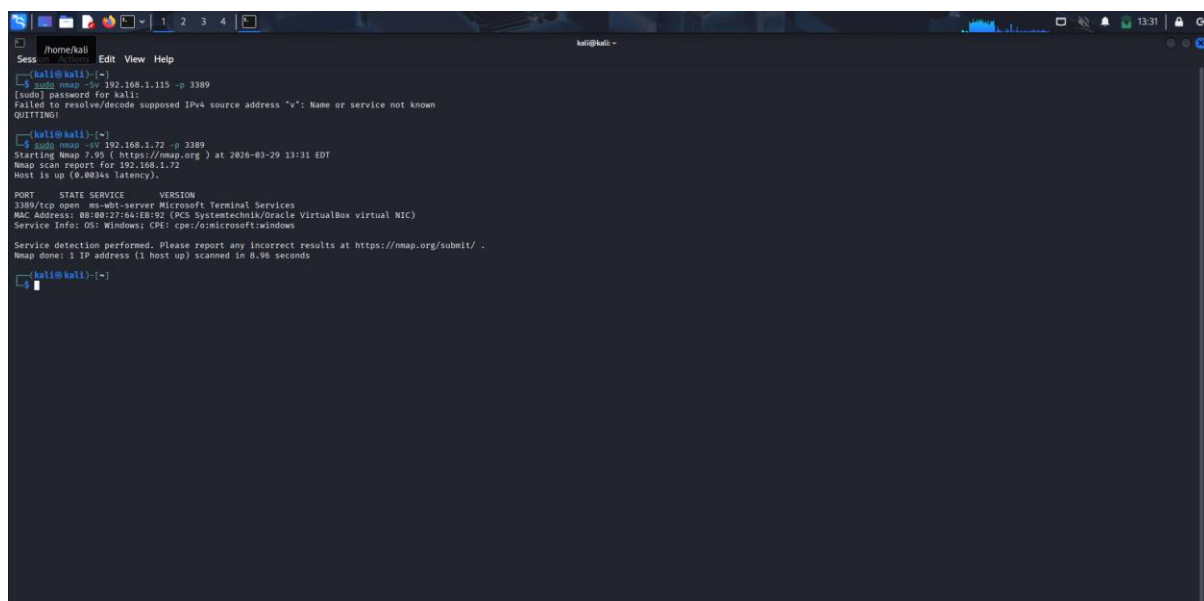
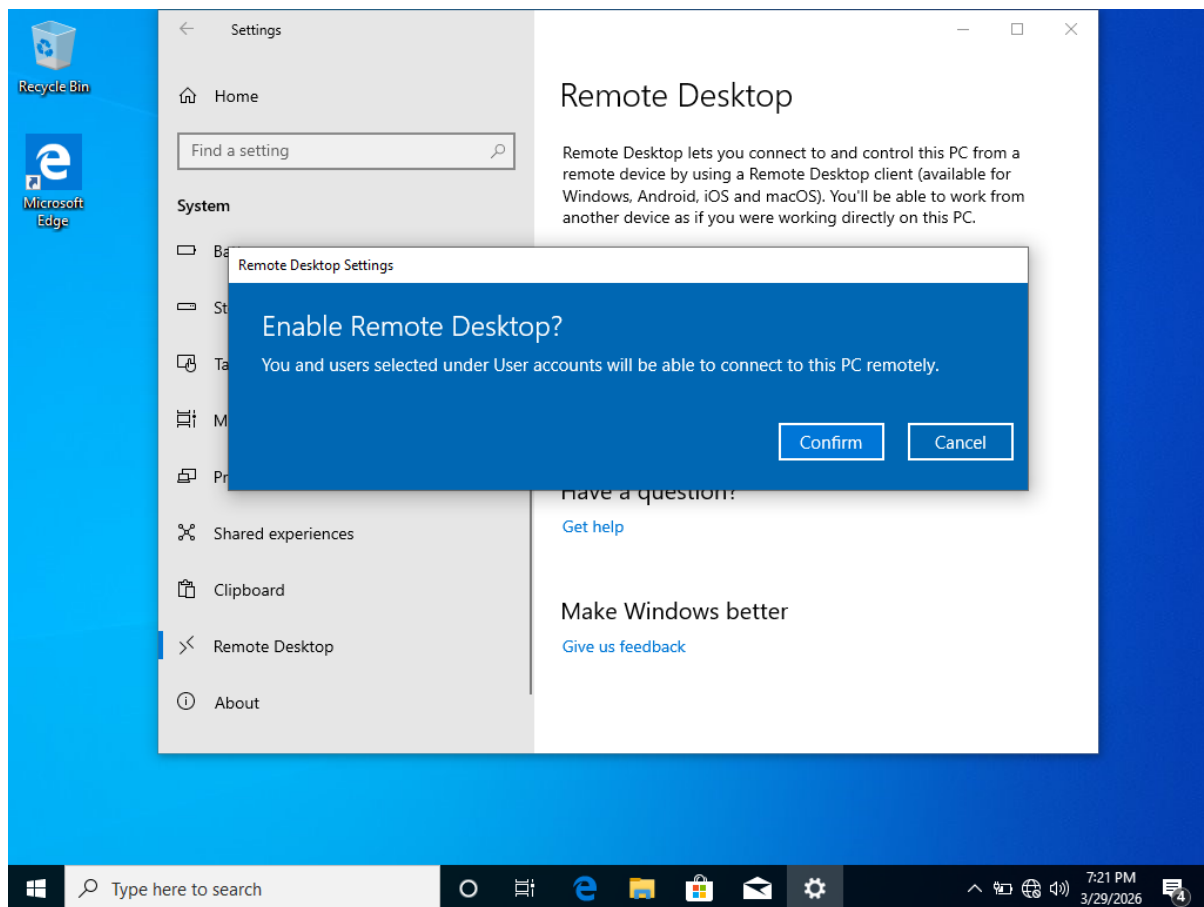


Per fare in modo che l'infrastruttura server-client sia in grado di inviare i log al siem è importante installare sulle vm l'Universal Forwarder di Splunk. Comuniceranno con l'ip 192.168.1.115



SIMULAZIONE DI UN EVENTO DI SICUREZZA, STUDIO ED INVIO DEI LOG AL SIEM

Una volta configurata tutta l'infrastruttura, dalla vm attaccante Kali si esegue una scansione con il tool nmap sul client windows 10. L'attacco scelto consiste nello sfruttare il servizio RDP sulla porta 3389, per poi verificare che il SIEM rilevi l'attacco.



La scansione con il tool nmap ci conferma che il servizio RDP è attivo sulla porta aperta 3389.

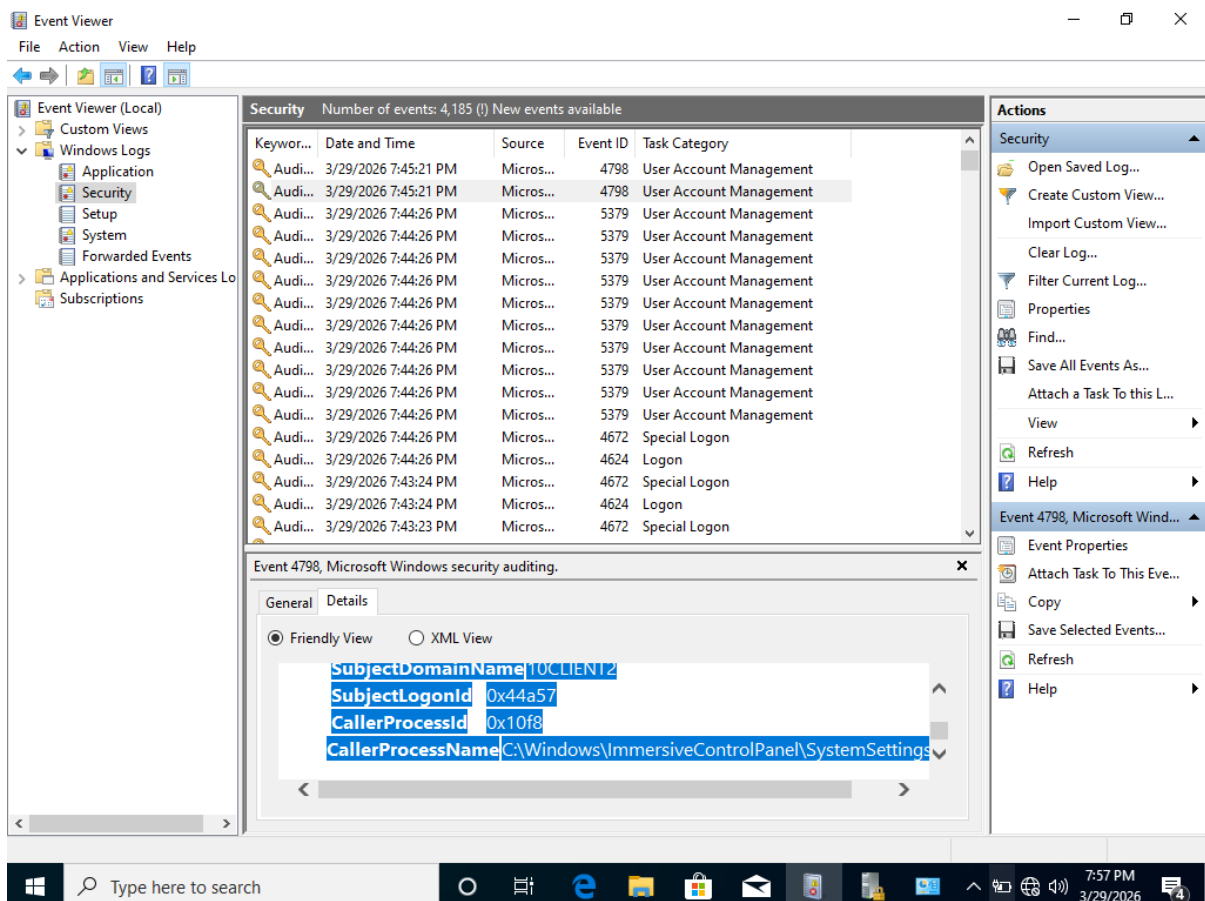
```
kali@kali: ~$ nmap -sV -p 3389 192.168.1.72
Nmap scan report for 192.168.1.72
Host is up (0.0013s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT      STATE SERVICE
3389/tcp   open  ms-rpc
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
|_ ssl-cert: Subject: commonName=10client2.lab.local
|_ Not valid before: 2026-03-28T17:22:34
|_ Not valid after: 2026-09-27T17:22:34
|_ rdp-ntlm-info:
|   Target Name: LAB
|   NetBIOS_Domain_Name: LAB
|   NetBIOS_Computer_Name: 10CLIENT2
|   DNS_Domain_Name: lab.local
|   DNS_Computer_Name: 10client2.lab.local
|   Product_Version: 10.0.18362
|_ System_Time: 2026-03-29T17:52:42+00:00
|_ Last-Date: 2026-03-29T17:52:42+00:00 - 39s from scanner time.
MAC Address: 08:00:27:64:EB:92 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Warning: OSscan results may be unreliable because we could not find at least 1 open and 1 closed port
Running (JUST GUESSING): Microsoft Windows 10[112019 (92%)]
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11 cpe:/o:microsoft:windows_server_2019
Aggressive OS guesses: Microsoft Windows 10 19H2 - 21H2 (92%), Microsoft Windows 11 (89%), Microsoft Windows 10 1809 (87%), Microsoft Windows 10 1909 (85%), Microsoft Windows 10 1909 - 20H2 (85%), Wind
wa Server 2019 (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ clock-skew: mean: -39s, deviation: 0s, median: -40s

TRACEROUTE
HOP RTT ADDRESS
1 1.33 ms 192.168.1.72

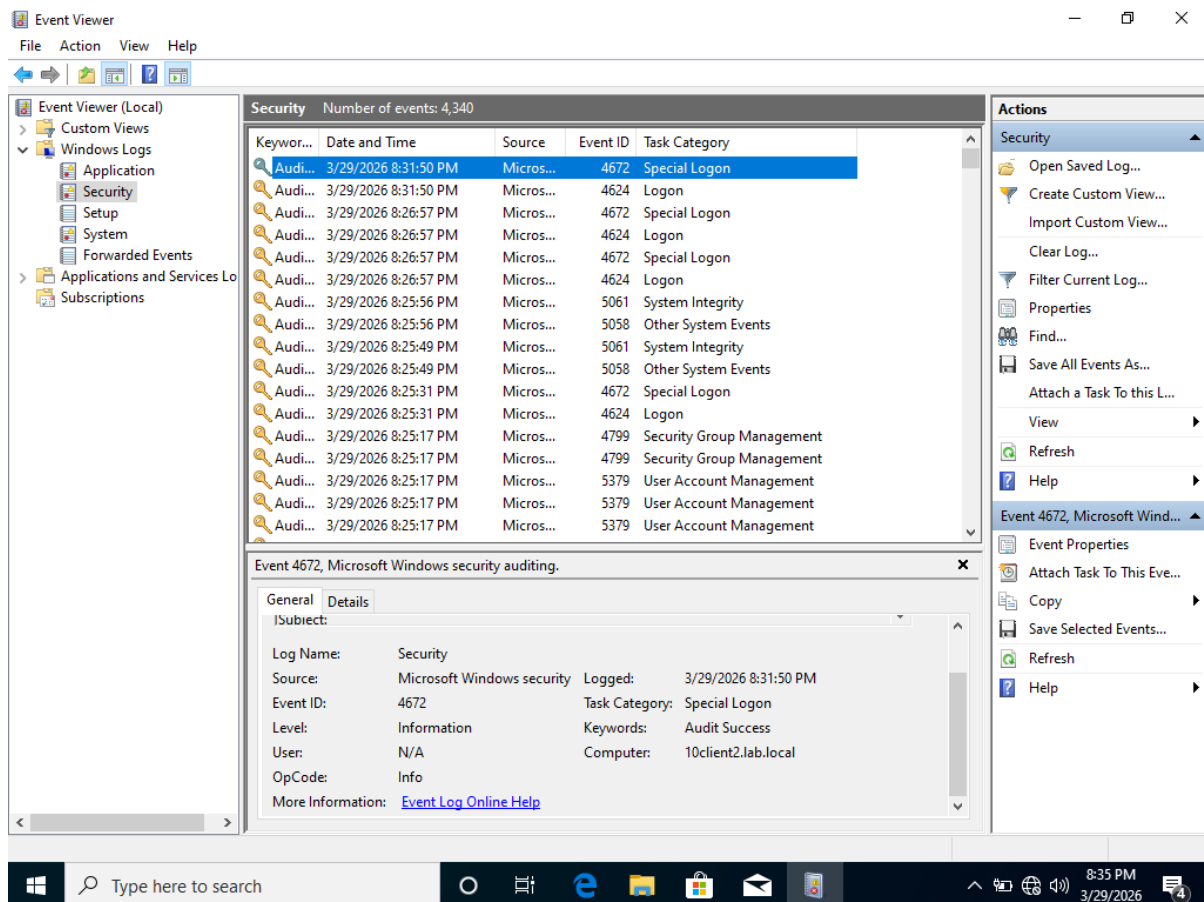
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.19 seconds
```

Successivo controllo dei log sul client windows 10



La scansione è individuata e le viene assegnato dal sistema l'ID 4789

Una seconda scansione è identificata dall'Event viewer nella sezione security con l'ID 4672



Si esegue successivamente un'altra scansione con il tool nmap ma questa volta sul server windows 2022

Recycle Bin

Administrator: Command Prompt

```
C:\Windows\system32>
C:\Windows\system32>eventcreate /t information /id 123 /l system d "Test Splunk Lab"
ERROR: Invalid syntax. 'infomration' value is not allowed for '/t' option.
Type "EVENTCREATE /?" for usage.

C:\Windows\system32>eventcreate /t information /id 123 /l system d "Test Splunk Lab"
ERROR: Invalid argument/option - 'd'.
Type "EVENTCREATE /?" for usage.

C:\Windows\system32>netsh advfirewall firewall add rule name="Allow Splunk Forwarder" dir=out action=allow program="C:\Program Files\SplunkUniversalForwarder\bin\splunkd.exe" enable=yes
Ok.

C:\Windows\system32>net stop SplunkForwarder
The SplunkForwarder service is not started.

More help is available by typing NET HELPMSG 3521.

C:\Windows\system32>net start SplunkForwarder
The SplunkForwarder service is starting....
The SplunkForwarder service was started successfully.

C:\Windows\system32>eventcreate /t information /id 999 /l system /d "Splunk Test Connection"
SUCCESS: An event of type 'information' was created with 'system' as the log.

C:\Windows\system32>
```

Type here to search

10:57 31/03/2026

Mar 31 10:56

Manage Indexes

Not Secure http://192.168.1.115:8000/en-US/manager/search/data/indexes?count=10&search=main

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Indexes

A repository for data in Splunk Enterprise. Indexes reside in flat files on the Splunk Enterprise instance known as the Indexer. [Learn more](#)

Showing 1-1 of 1 Indexes

main

Name	Status	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path
main	Active	Events	system	1 MB	488.28 GB	0	—	—	\$SPLUNK_DB/defaultdb/db	—

Showing 1-1 of 1 Indexes

Trash

Si effettua una ulteriore prova lanciando una nuova scansione sul client windows10 dalla vm kali

```

kali@kali: ~$ sudo nmap -sV -O 192.168.1.72
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-31 05:07 EDT
Nmap scan report for 192.168.1.72
Host is up (0.0016s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc      Microsoft Windows RPC
239/tcp    open  netbios-ssn Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ms-wbt-server Microsoft Terminal Services
|_ tcp-ssl-info:
|   Target_Name: LAB
|   NetBIOS_Domain_Name: LAB
|   NetBIOS_Computer_Name: 10CLIENT2
|   DNS_Domain_Name: lab.local
|   DNS_Computer_Name: 10client2.lab.local
|   DNS_Tree_Name: lab.local
|   Product_Version: 10.0.18362
|_ System_Time: 2026-03-31T09:07:18+00:00
ssl-date: 2026-03-31T09:07:22+00:00; -1m00s from scanner time.
ssl-cert: Subject: commonName=10client2.lab.local
Not valid before: 2026-03-28T17:22:34
Not valid after: 2026-09-27T17:22:34
MAC Address: 08:00:27:64:EB:92 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Microsoft Windows 10
OS CPE: cpe:/o:microsoft:windows_10
OS details: Microsoft Windows 10 1709 - 21H2
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_ clock skew: mean: -58s, deviation: 1s, median: -58s
|_ smb2-security-mode:
|   3.1.1:
|     Message signing enabled but not required
|_ smb2-time:
|   date: 2026-03-31T09:07:18
|   start_date: N/A
|_ nbstat: NetBIOS name: 10CLIENT2, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:64:eb:92 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

TRACEROUTE
HOP RTT ADDRESS
0 1.65 ms 192.168.1.72

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 22.24 seconds

kali@kali: ~$

```

Dopo alcuni tentativi da Kali si ottiene l'accesso all'interfaccia del client

```

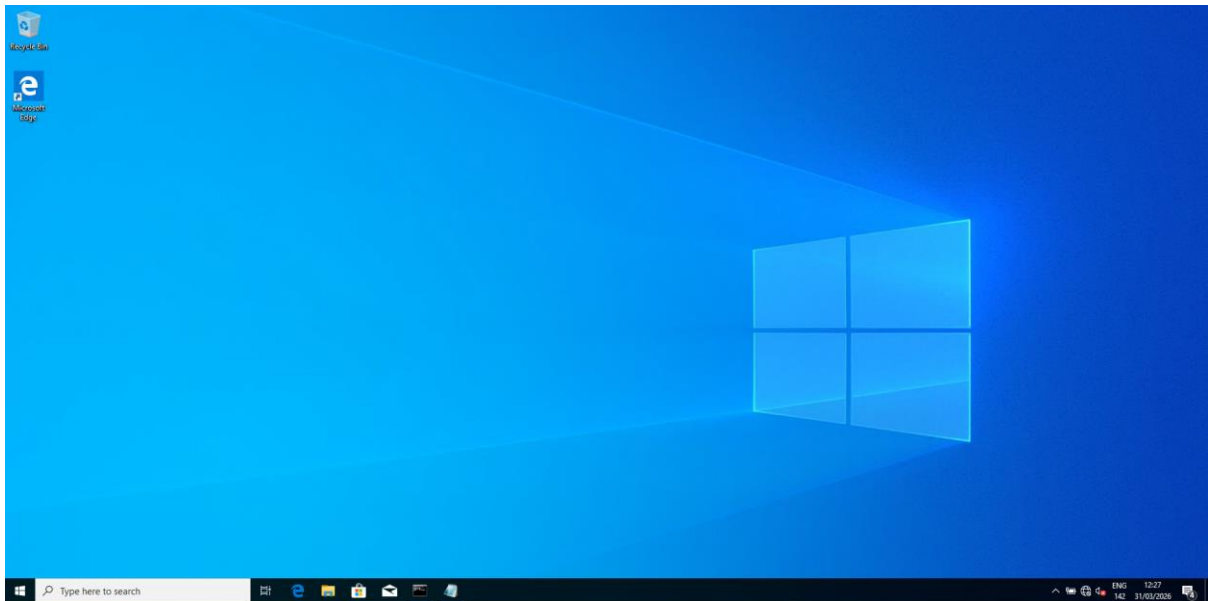
[06:23:11:599] [a132a:0000a170] [ERROR][com.freerdp.core.connection] [rdp_client_wait_for_activation] Timeout waiting for activation
[06:23:11:599] [a132a:0000a170] [ERROR][com.freerdp.core] [rdp_client_wait_for_activation] ERRCORE_ACTIVATION_TIMEOUT [0-0000201c]

kali@kali:~$
kali@kali:~$ ./freerdp -i 192.168.1.72 /u:vbuser /p:kali1234 /f /c:certificate -c:clipboard /dynamic-resolution
[06:26:15:353] [a1928:0000a3b8] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Using /p is insecure
[06:26:15:353] [a1928:0000a3b8] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Passing credentials or secrets via command line might expose these in the process list
[06:26:15:353] [a1928:0000a3b8] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Consider using one of the following (more secure) alternatives:
[06:26:15:353] [a1928:0000a3b8] [- warn_credential_args] - /args- from pipe in arguments from stdin, file or file descriptor
[06:26:15:353] [a1928:0000a3b8] [- warn_credential_args] - /from-stdin pass the credential via stdin
[06:26:15:353] [a1928:0000a3b8] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] - set environment variable FREERDP_ASKPASS to have a gui tool query for credentials
[06:26:15:380] [a1928:0000a3b8] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] - /load_map from xRDPfile
[06:26:15:380] [a1928:0000a3b8] [WARN][com.freerdp.client.x11] [load_map_from_xRDPfile] : keycode: 0x5d -> no RDP scancode found
[06:26:15:451] [a1928:0000a3b8] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] Certificate not checked, /c:certificate in use
[06:26:15:451] [a1928:0000a3b8] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] This prevents MITM attacks from being detected!
[06:26:15:452] [a1928:0000a3b8] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] Avoid using this unless in a secure LAN (no internet) environment
[06:26:15:452] [a1928:0000a3b8] [ERROR][com.wingapi.kerberos] [- kerberos_AcquireCredentialHandle] krb5_name_name (Configuration file does not specify default realm [-1765328160])
[06:26:15:452] [a1928:0000a3b8] [ERROR][com.wingapi.kerberos] [- kerberos_AcquireCredentialHandle] krb5_data_name (Configuration file does not specify default realm [-1765328160])
[06:26:15:462] [a1928:0000a3b8] [ERROR][com.freerdp.gdi] [- nla_recv_pdu] ERRCORENET_LOGON_FAILURE [0-0000201c]
[06:26:15:462] [a1928:0000a3b8] [ERROR][com.freerdp.core.rdp] [rdp_recv_callback_int] [0-55807ec9ab]: CONNECTION_STATE_NLA_nla_recv_pdu() fail
[06:26:15:462] [a1928:0000a3b8] [ERROR][com.freerdp.core.rdp] [rdp_recv_callback_int] [0-55807ec9ab]: CONNECTION_STATE_NLA_state_RUN_FAILED [-1]
[06:26:15:462] [a1928:0000a3b8] [ERROR][com.freerdp.core.transport] [transport_check_fds] transport_check_fds: transport=ReceiveCallback() - STATE_RUN_FAILED [-1]

kali@kali:~$
kali@kali:~$ ./freerdp -i 192.168.1.72 /u:vbuser /p:kali1234 /f /c:certificate -c:clipboard /dynamic-resolution
[06:26:44:270] [a1262:0000a0b2] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Using /p is insecure
[06:26:44:270] [a1262:0000a0b2] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Passing credentials or secrets via command line might expose these in the process list
[06:26:44:270] [a1262:0000a0b2] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] Consider using one of the following (more secure) alternatives:
[06:26:44:270] [a1262:0000a0b2] [- warn_credential_args] - /args- from pipe in arguments from stdin, file or file descriptor
[06:26:44:270] [a1262:0000a0b2] [- warn_credential_args] - /from-stdin pass the credential via stdin
[06:26:44:270] [a1262:0000a0b2] [WARN][com.freerdp.client.common.cmdline] [- warn_credential_args] - set environment variable FREERDP_ASKPASS to have a gui tool query for credentials
[06:26:44:324] [a1262:0000a0b2] [WARN][com.freerdp.client.x11] [load_map_from_xRDPfile] : keycode: 0x5d -> no RDP scancode found
[06:26:44:324] [a1262:0000a0b2] [WARN][com.freerdp.client.x11] [load_map_from_xRDPfile] : keycode: 0x5d -> no RDP scancode found
[06:26:44:393] [a1262:0000a0b2] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] Certificate not checked, /c:certificate in use
[06:26:44:393] [a1262:0000a0b2] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] This prevents MITM attacks from being detected!
[06:26:44:393] [a1262:0000a0b2] [WARN][com.freerdp.crypto] [- tls_verify_certificate] [DANGER] Avoid using this unless in a secure LAN (no internet) environment
[06:26:44:393] [a1262:0000a0b2] [ERROR][com.wingapi.kerberos] [- kerberos_AcquireCredentialHandle] krb5_name_name (Configuration file does not specify default realm [-1765328160])
[06:26:44:393] [a1262:0000a0b2] [ERROR][com.wingapi.kerberos] [- kerberos_AcquireCredentialHandle] krb5_data_name (Configuration file does not specify default realm [-1765328160])
[06:26:45:211] [a1262:0000a0b2] [INFO][com.freerdp.gdi] [- gdi_int_init] Local framebuffer format PIXEL_FORMAT_BGRX32
[06:26:45:211] [a1262:0000a0b2] [INFO][com.freerdp.gdi] [- gdi_int_init] Remote framebuffer format PIXEL_FORMAT_BGRX32
fuse: warning: library too old,
some operations may not work
[06:26:45:273] [a1262:0000a0b2] [INFO][com.freerdp.channels.rdpndp.client] [rdpsnd_load_device_plugin] [static] loaded fake backend for rdpndp
[06:26:45:273] [a1262:0000a0b2] [INFO][com.freerdp.channels.rdpndp.client] [rdpsnd_load_device_plugin] [static] loaded fake backend for rdpndp
[06:26:45:274] [a1262:0000a0b2] [INFO][com.freerdp.channels.drdynvc.client] [rdvdm_load_addin] Loading Dynamic Virtual Channel alsupfx
[06:26:45:274] [a1262:0000a0b2] [INFO][com.freerdp.channels.drdynvc.client] [rdvdm_load_addin] Loading Dynamic Virtual Channel rdpfx
[06:26:45:274] [a1262:0000a0b2] [INFO][com.freerdp.channels.drdynvc.client] [rdvdm_load_addin] Loading Dynamic Virtual Channel rdpndp
[06:26:45:274] [a1262:0000a0b2] [INFO][com.freerdp.channels.drdynvc.client] [rdvdm_load_addin] Loading Dynamic Virtual Channel rdpndp
[06:26:47:080] [a1262:0000a0b2] [INFO][com.freerdp.channels.rdpndp.client] [rdpsnd_load_device_plugin] [dynamic] loaded fake backend for rdpndp
[06:26:47:080] [a1262:0000a0b2] [INFO][com.freerdp.channels.rdpndp.client] [rdpsnd_load_device_plugin] [dynamic] loaded fake backend for rdpndp
[06:29:20:877] [a1262:0000a0b2] [INFO][com.freerdp.core] [rdp_print_errinfo] ERRCORE_RPC_INITIATED_DISCONNECT [0-00000001] The disconnection was initiated by an administrative tool on the server in another session.
[06:29:20:877] [a1262:0000a0b2] [ERROR][com.freerdp.core] [rdp_set_error_info] ERRCORE_RPC_INITIATED_DISCONNECT [0-00000001]

kali@kali:~$

```



Event Viewer

File Action View Help

Event Viewer (Local)

- Custom Views
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Logs
- Subscriptions

Security Number of events: 6,680 (1) New events available

Keyword...	Date and Time	Source	Event ID	Task Category
Audi...	31/03/2026 12:35:22	Micros...	4634	Logoff
Audi...	31/03/2026 12:35:22	Micros...	4634	Logoff
Audi...	31/03/2026 12:35:20	Micros...	4634	Logoff
Audi...	31/03/2026 12:35:19	Micros...	4634	Logoff
Audi...	31/03/2026 12:35:19	Micros...	4634	Logoff
Audi...	31/03/2026 12:35:17	Micros...	4778	Other Logon/Logoff Events
Audi...	31/03/2026 12:35:16	Micros...	4672	Special Logon
Audi...	31/03/2026 12:35:16	Micros...	4672	Special Logon
Audi...	31/03/2026 12:35:16	Micros...	4627	Group Membership
Audi...	31/03/2026 12:35:16	Micros...	4624	Logon
Audi...	31/03/2026 12:35:16	Micros...	4627	Group Membership
Audi...	31/03/2026 12:35:16	Micros...	4624	Logon
Audi...	31/03/2026 12:35:16	Micros...	4648	Logon
Audi...	31/03/2026 12:35:16	Micros...	4800	Other Logon/Logoff Events
Audi...	31/03/2026 12:35:16	Micros...	4627	Group Membership
Audi...	31/03/2026 12:35:16	Micros...	4624	Logon
Audi...	31/03/2026 12:35:16	Micros...	4648	Logon

Event 4634, Microsoft Windows security auditing.

General Details

An account was logged off.

Subject:

Log Name: Security

Source: Microsoft Windows security

Event ID: 4634

Level: Information

Logged: 31/03/2026 12:35:22

Task Category: Logoff

Keywords: Audit Success

Actions

- Security
- Open Saved Log...
- Create Custom View...
- Import Custom View...
- Clear Log...
- Filter Current Log...
- Properties
- Find...
- Save All Events As...
- Attach a Task To this L...
- View
- Refresh
- Help
- Event 4634, Microsoft Wind...
- Event Properties
- Attach Task To This Eve...
- Copy
- Save Selected Events...
- Refresh
- Help

Evidenza dei log relativi all'accesso tramite RDP con l'utilizzo della query: index=main
192.168.1.116 sul client windows10

The screenshot shows a web browser window displaying the Splunk search results interface. The address bar shows the URL: `http://192.168.1.115:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Amain`. The search results are displayed in a table with columns: **Time** and **Event**. The **Time** column shows the timestamp: `2026-03-31T15:21:29.759+02:00`. The **Event** column contains a detailed log entry from `WinEventLog:Security`. The entry details include:

- Account_Domain**: 10CLIENT2
- Account_Name**: vboxuser
- Client_Address**: 192.168.1.116
- Client_Name**: kali
- ComputerName**: 10client2.lab.local
- EventCode**: 4778
- EventType**: 0
- Keywords**: Audit Success
- LogName**: Security
- Logon_ID**: 0x3B756
- Message**: A session was reconnected to a Window Station. Subject: Account Name: vboxuser Account Domain: 10CLIENT2 Logon ID: 0x3B756 Session: Session Name: RDP-Tcp#0 Additional Information: Client Name: kali Client Address: 192.168.1.116 This event is generated when a user reconnects to an existing Terminal Services session, or when a user switches to an existing desktop using Fast User Switching.
- OpCode**: Info
- RecordNumber**: 8843
- Session_Name**: RDP-Tcp#0
- SourceName**: Microsoft Windows security auditing.
- TaskCategory**: Other Logon/Logoff Events
- Type**: Information

Search | Splunk 10.2.1 x Passwords

Not Secure http://192.168.1.115:8000/en-US/app/search/search?earliest=0&latest=&q=search index%

✓ 26 events (before 4/1/26 12:51:38.000 PM) No Event Sampling

Events (26) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Terminal

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- Account_Domain 3
- Account_Name 4
- Authentication_Package 2
- Client_Address 1
- Client_Name 1
- ComputerName 1
- Elevated_Token 2
- EventCode 5
- EventType 1
- Impersonation_Level 1
- Index 1
- Key_Length 2
- Keywords 2
- linecount 5

i	Time	Event
>	3/31/26	03/31/2026 03:28:33.496 PM
	3:28:33.496 PM	... 21 lines omitted ...
		Client Name: kali
		Client Address: 192.168.1.116
		Show all 27 lines
		host = 10client2 source = WinEventLog:Security sourcetype = WinEventLog:Security
>	3/31/26	03/31/2026 03:23:46.971 PM
	3:23:46.971 PM	... 21 lines omitted ...
		Client Name: kali
		Client Address: 192.168.1.116
		This event is generated when a user reconnects to an existing Terminal Services session, or when a user switches to an existing desktop using Fast User Switching.
		Show all 26 lines
		host = 10client2 source = WinEventLog:Security sourcetype = WinEventLog:Security
>	3/31/26	03/31/2026 03:23:45.708 PM
	3:23:45.708 PM	... 41 lines omitted ...

Log relativi all'host server a seguito di alcuni tentativi di accesso dalla vm kali tramite l'utilizzo del tool hydra sfruttando il protocollo rdp. I parametri passati ad hydra sono lo user *Valeria* e la wordlist *rockyou*.

```
(kali@kali)~$ hydra -l valeria -P /usr/share/wordlists/rockyou.txt 192.168.1.70 rdp -t1
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-01 10:56:03
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[WARNING] Restorefile (you have 10 seconds to abort... (use option -i to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 1 task per 1 server, overall 1 task, 14344399 login tries (1:1/p:14344399), ~14344399 tries per task
[DATA] attacking rdp://192.168.1.70:3389/
[STATUS] 50.00 tries/min, 50 tries in 00:01h, 14344319 to do in 4781:37h, 1 active
[STATUS] 50.67 tries/min, 152 tries in 00:03h, 14344267 to do in 4718:31h, 1 active
[STATUS] 54.29 tries/min, 388 tries in 00:07h, 14344019 to do in 4403:52h, 1 active
```

Apr 1 17:13

Search | Splunk 10.2.1 x Passwords

Not Secure http://192.168.1.116:8000/en-US/app/search/search?earliest=1774908000&latest=1775000000

index=main 192.168.1.116 Time range: from Mar 31 through Apr ...

1,088 events (3/31/26 12:00:00.000 AM to 4/2/26 12:00:00.000 AM) No Event Sampling Job

Events (1,088) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect 1 hour per column

Format Show: 20 Per Page View: List Prev 1 2 3 4 5 6 7 8 Next

VBox_GAs_7.2.2 All Fields

SELECTED FIELDS

- host 2
- source 1
- sourcetype 2

INTERESTING FIELDS

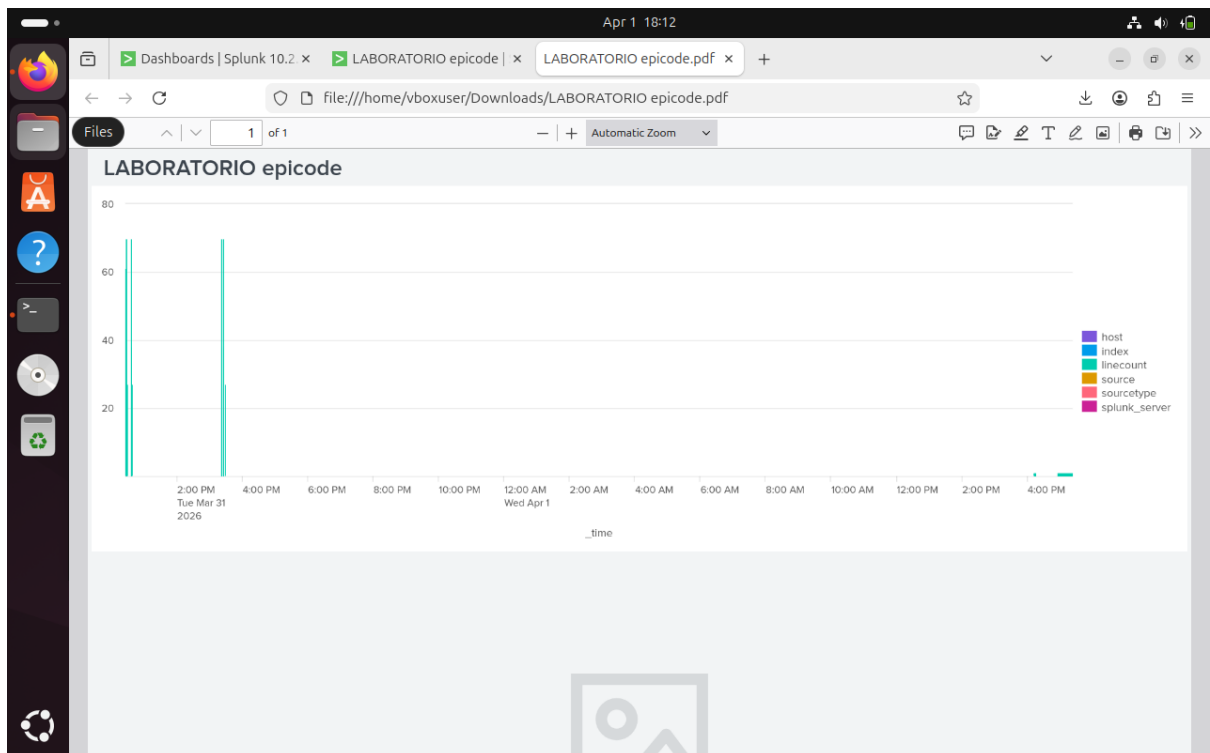
- index 1
- linecount 6
- punct 5
- splunk_server 1

46 more fields

+ Extract New Fields

i	Time	Event
>	4/1/26 5:12:19.714 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4625</EventID><Version>0</Version><Level>0</Level><Task>12544</Task><Opcode>0</Opcode><Keywords>0x8010000000000000</Keywords><TimeCreated SystemTime='2026-04-01T15:12:19.7141853Z' /><EventRecordID>20455</EventRecordID><Correlation><Execution ProcessID='636' ThreadID='4972' /><Channel>Security</Channel><Computer>SERVER.lab.local</Computer><Security /></System><EventData><Data Name='SubjectUserSid'>S-1-0-0</Data><Data Name='SubjectUserName'></Data><Data Name='SubjectDomainName'></Data><Data Name='SubjectLogonId'>0x0</Data><Data Name='TargetUserSid'>S-1-0-0</Data><Data Name='TargetUserName'>valeria</Data><Data Name='TargetDomainName'></Data><Data Name='Status'>0xc000006d</Data><Data Name='FailureReason'>%2313</Data><Data Name='SubStatus'>0xc000006a</Data><Data Name='LogonType'>3</Data><Data Name='LogonProcessName'>NtLmSsp </Data><Data Name='AuthenticationPackageName'>NTLM</Data><Data Name='WorkstationName'>kali</Data><Data Name='TransmittedServices'></Data><Data Name='LmPackageName'></Data><Data Name='KeyLength'>0</Data><Data Name='ProcessId'>0x0</Data><Data Name='ProcessName'></Data><Data Name='IpAddress'>192.168.1.116</Data><Data Name='IpPort'>0</Data></EventData></Event>
>	4/1/26 5:12:18.859 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Security-Auditing' Guid='{54849625-5478-4994-a5ba-3e3b0328c30d}' /><EventID>4625</EventID><Version>0</Version><Level>0</Level><Task>12544</Task><Opcode>0</Opcode><Keywords>0x8010000000000000</Keywords><TimeCreated SystemTime='2026-04-01T15:12:18.8597646Z' /><EventRecordID>20454</EventRecordID><Correlation><Execution ProcessID='636' ThreadID='4972' /><Channel>Security</Channel><Computer>SERVER.lab.local</Computer><Security /></System><EventData><Data Name='SubjectUserSid'>S-1-0-0</Data><Data Name='SubjectUserName'></Data><Data Name='SubjectDomainName'></Data><Data Name='SubjectLogonId'>0x0</Data><Data Name='TargetUserSid'>S-1-0-0</Data><Data Name='TargetUserName'>valeria</Data><Data Name='TargetDomainName'></Data><Data Name='Status'>0xc000006d</Data><Data Name='FailureReason'>%2313</Data><Data Name='SubStatus'>0xc000006a</Data><Data Name='LogonType'>3</Data><Data Name='LogonProcessName'>NtLmSsp </Data><Data Name='AuthenticationPackageName'>NTLM</Data><Data Name='WorkstationName'>kali</Data><Data Name='TransmittedServices'></Data><Data Name='LmPackageName'></Data><Data Name='KeyLength'>0</Data><Data Name='ProcessId'>0x0</Data><Data Name='ProcessName'></Data><Data Name='IpAddress'>192.168.1.116</Data><Data Name='IpPort'>0</Data></EventData></Event>

host = SERVER source = WinEventLog:Security sourcetype = XmlWinEventLog:Security



Il laboratorio correttamente configurato consente di prendere visione dei log che il client e il server inoltrano al siem splunk sulla vm ubuntu a seguito di alcuni tentativi di scansione e attacchi di bruteforce inviati dalla vm kali.

FINE PROGETTO 01/04/2026